OPERATING SYSTEMS

Storage Management

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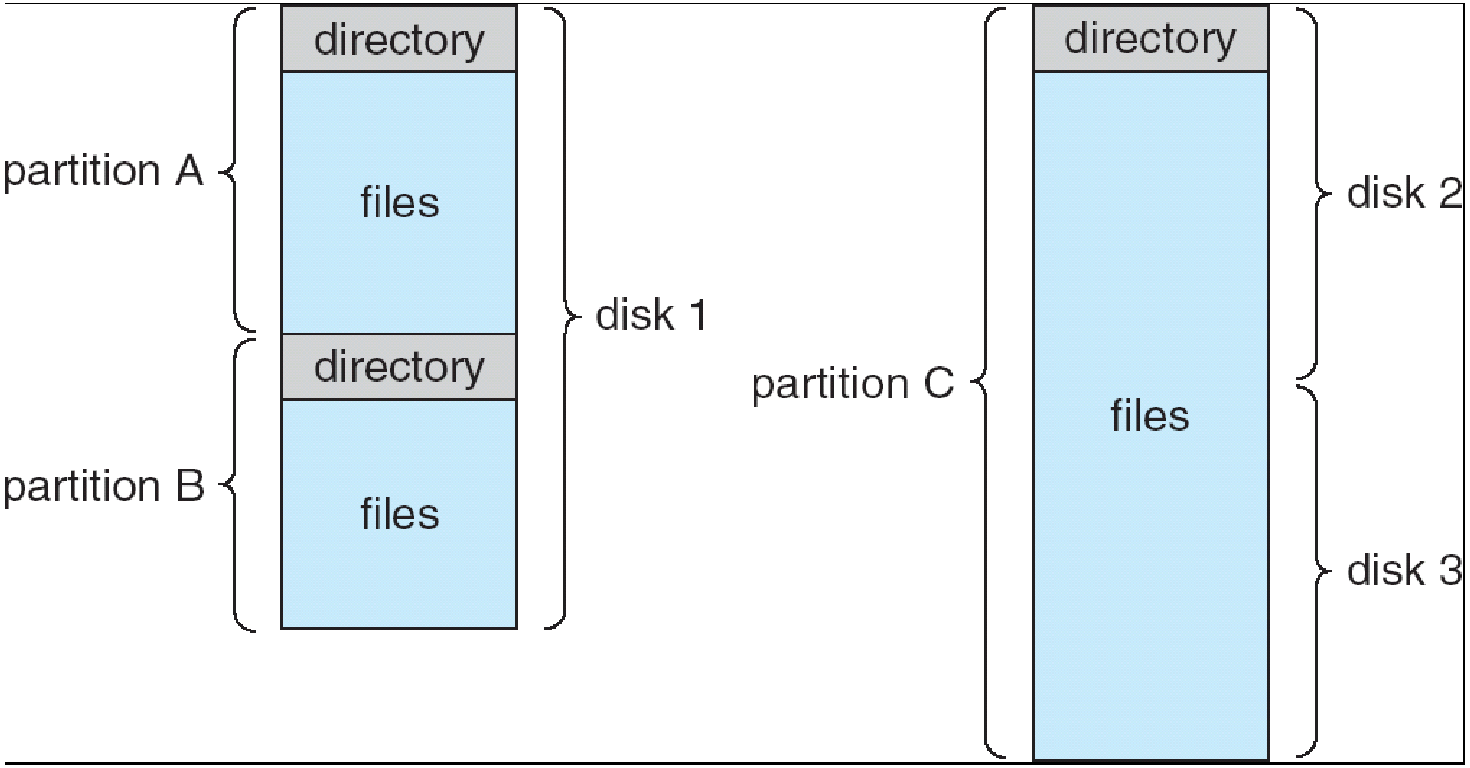
FS on disk

- Could use entire disk space for a FS, but
  - A system could have multiple FSes
  - Want to use some disk space for swap space
- Disk divided into partitions, slices or minidisks
  - Chunk of storage that holds a FS is a volume
  - Directory structure maintains info of all files in the volume
    - Name, location, size, type, …

Directory Structure



Directories

Directories/folders keep track of files

Is a symbol table that translates file names as directory entries

Directory itself is a file

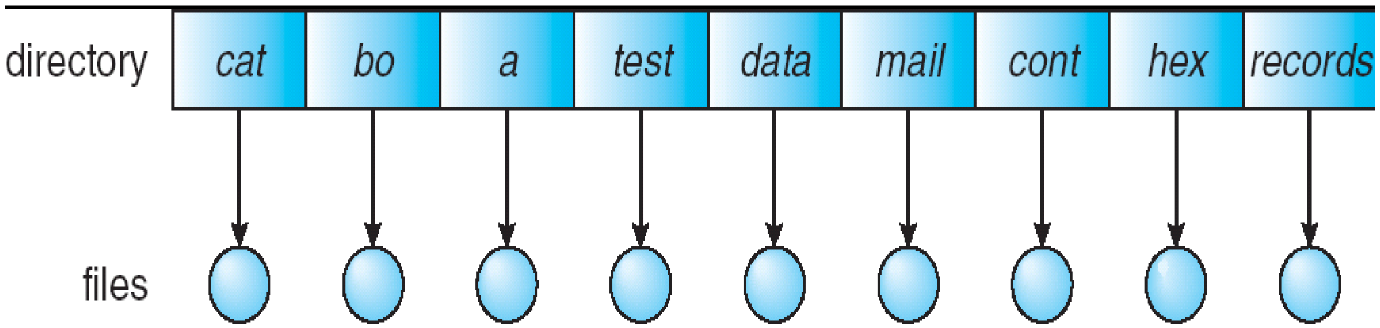
Operations:

- Search a file
- Create a file
- Delete a file
- List directory
- Rename a file
- Traversing the FS

Single Level Directory

One directory for all files in the volume

Called root directory



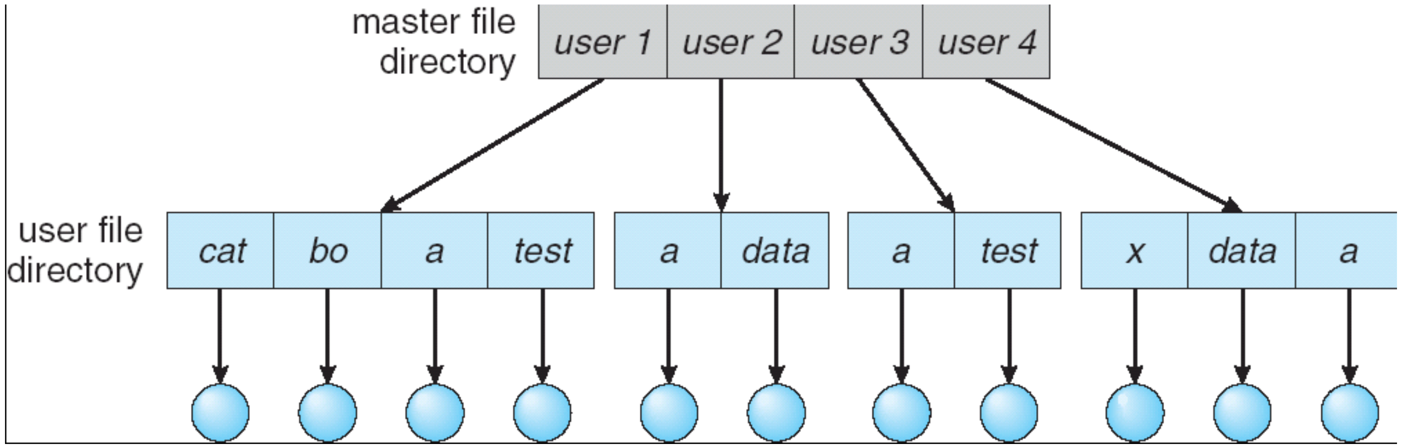
Used in early PCs, even the first supercomputer CDC 6600

Pros: simplicity, ability to quickly locate files

Cons: inconvenient naming (uniqueness, remembering all)

Two level Directory

Each user has a separate directory



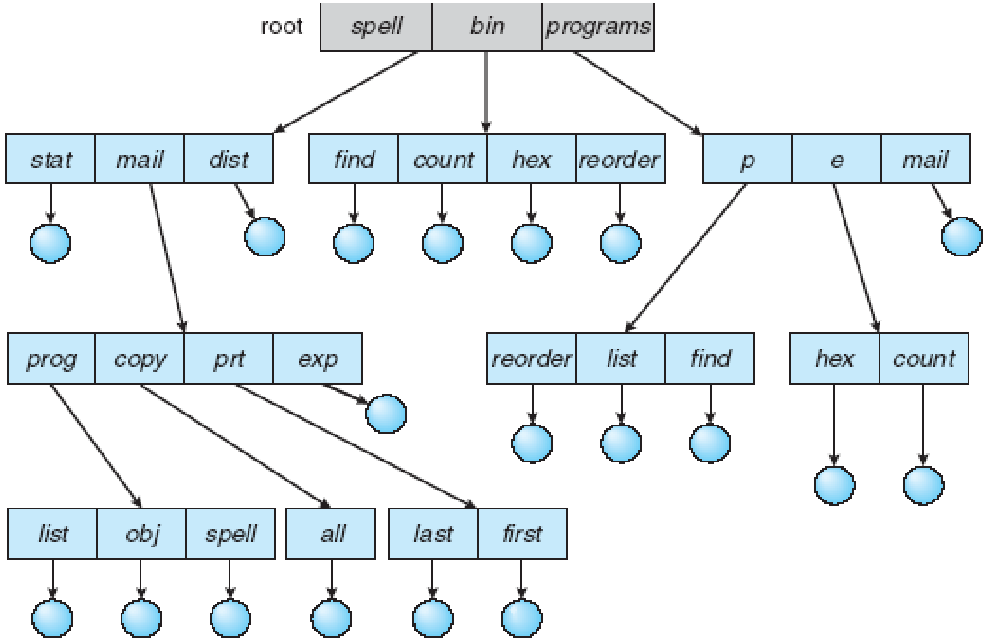
Solves name collision, but what if user has lots of files

May not allow a user to access other users' files

Tree Structured Directory

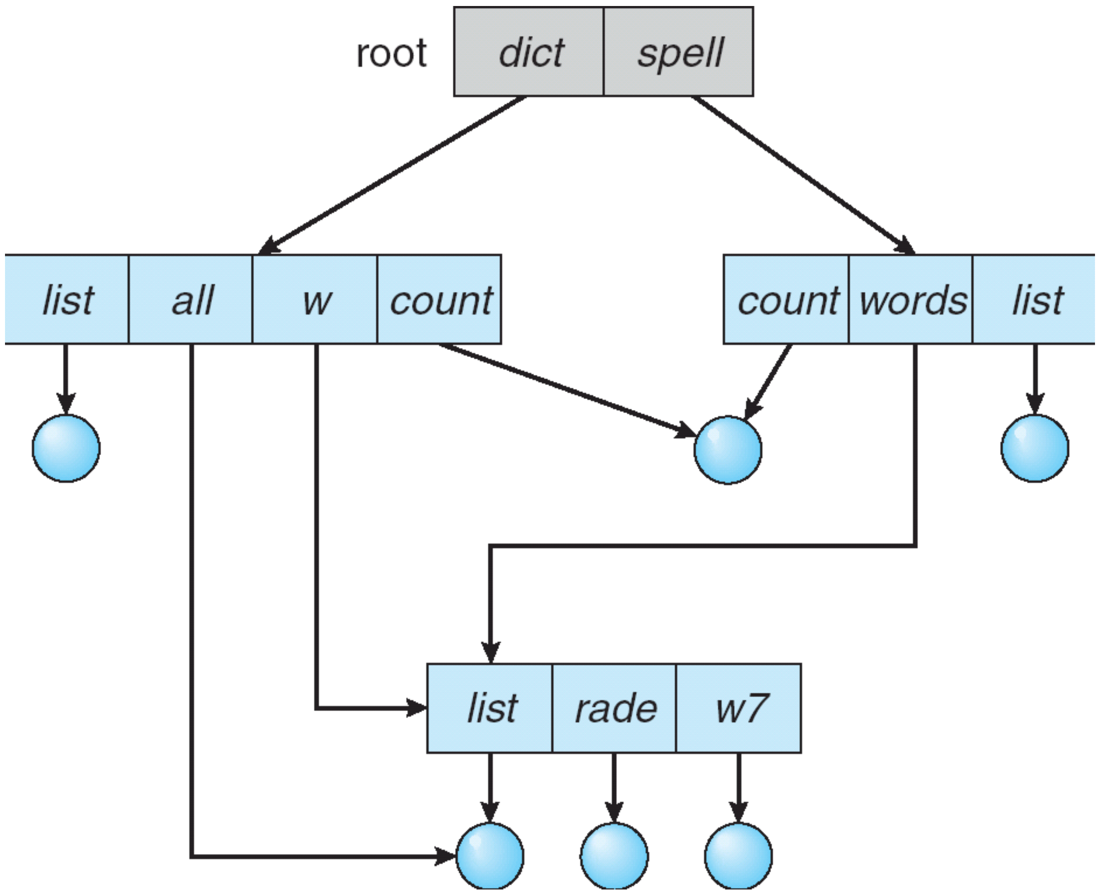
Directory is now a tree of arbitrary height

- Directory contains files and subdirectories
- A bit in directory entry differentiates files from subdirectories



Acyclic Graph Directories

Shares subdirectories and files



Directory Access

To access a file, the user should either:

- Go to the directory where file resides, or
- Specify the **path** where the file is

Path names are either absolute or relative

- Absolute: path of file from the root directory
- Relative: path from the current working directory

Most OSes have two special entries in each directory:

- "." for current directory and ".." for parent

THANK YOU

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